Status	Finished
Started	Saturday, 17 October 2026, 4:40 PM
Completed	Saturday, 17 October 2026, 4:48 PM
Duration	8 mins 25 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

b. Must use a sorted array

c. There must be a mechanism to access middle element directly

d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **2**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the right sublist

b. The list is sorted

c. The search ends successfully ✓

d. The search continues in the left sublist

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element

b. By dividing the list length by two ✓

c. By using a hash function

d. By comparing each element sequentially

Question **4**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is small and sorted
c. When the list is small and unsorted ✓
d. When the list is large and unsorted

Question **5**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all
b. When the item is the last element in the array
c. Item is the last element in the array or item is not there at all
d. When the item is somewhere in the middle of the array ✓

Question **6**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The middle element is chosen randomly
b. The search stops
c. The lower middle element is chosen as the middle element ✗
d. The higher middle element is chosen as the middle element

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The list is sorted
b. The search stops
c. The search continues in the left or right sublist ✓
d. The search continues from the beginning

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.
b. Linear search divides the list into halves.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search is more efficient than binary search on large lists.

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O👎 ✓
b. O(n log n)
c. O(log n)
d. O(1)

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Dynamic search
b. Linear search ✓
c. Binary search
d. Hashing

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓
b. In the right sublist
c. In the entire list
d. At the beginning of the list

Question **12**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the first element in the list
b. Sort the list ✗
c. Compare the target element with the middle element in the list
d. Divide the list into two equal parts

Question **13**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)
b. Linear search ✓
c. Hash search
d. Binary search

Question **14**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 99 ✓
b. 94 and 99
c. 90 and 100
d. 89 and 94

Question **15**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Hash search
b. Binary search
c. Interpolation search
d. Linear search ✓